Navigating the Realtime Spectrum

From Eventual Consistency to Sub-Millisecond Peer-to-Peer Architecture

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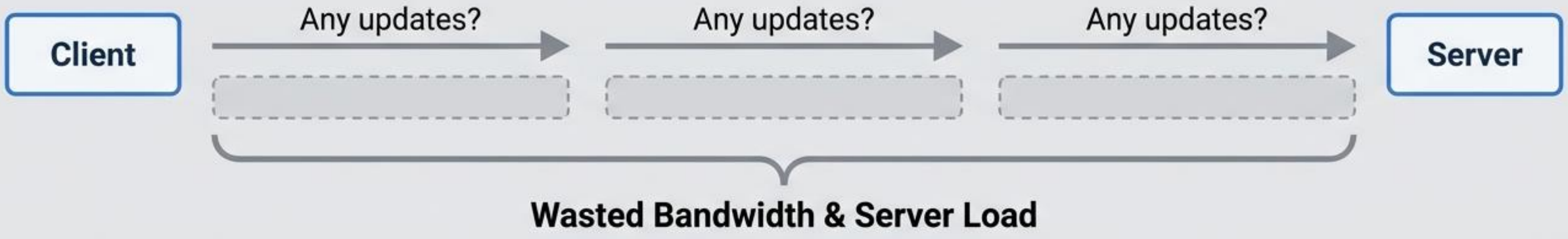
The Inefficiency of the Polling Tax

Standard HTTP request/response architectures force a compromise between stale UIs and wasted bandwidth.

The Problem: Polling



The Solution: Event-Driven

Realtime architecture eliminates the polling tax, enabling true collaboration without the overhead.

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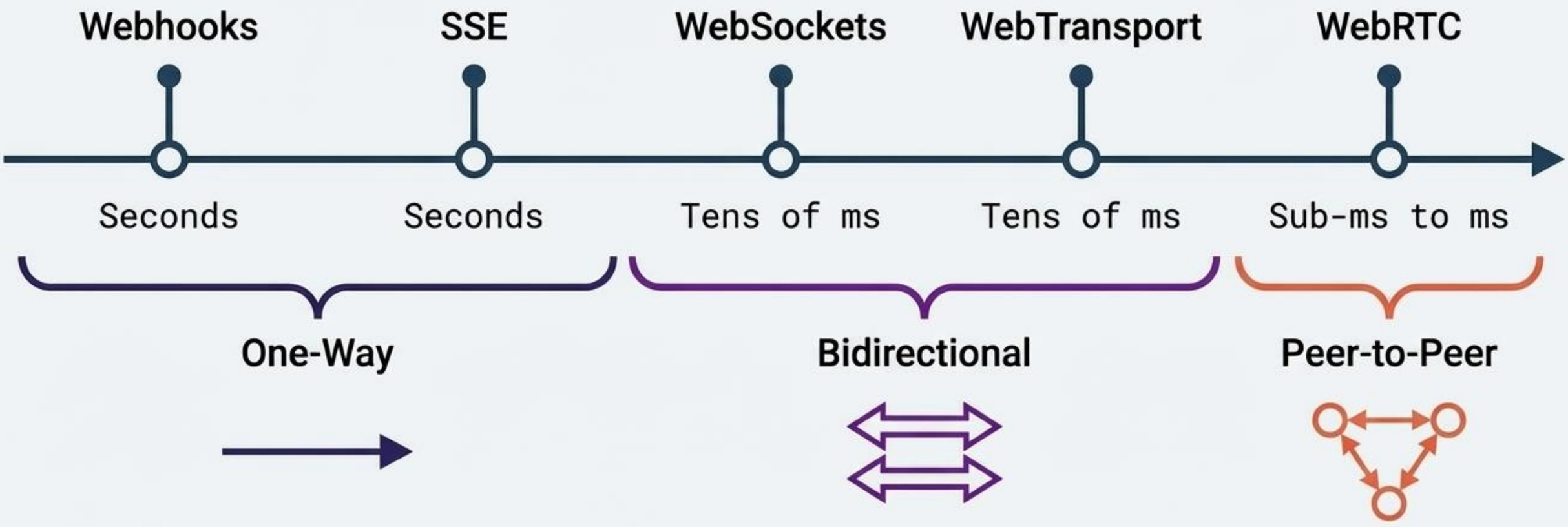
Realtime is a Spectrum, Not a Single Standard



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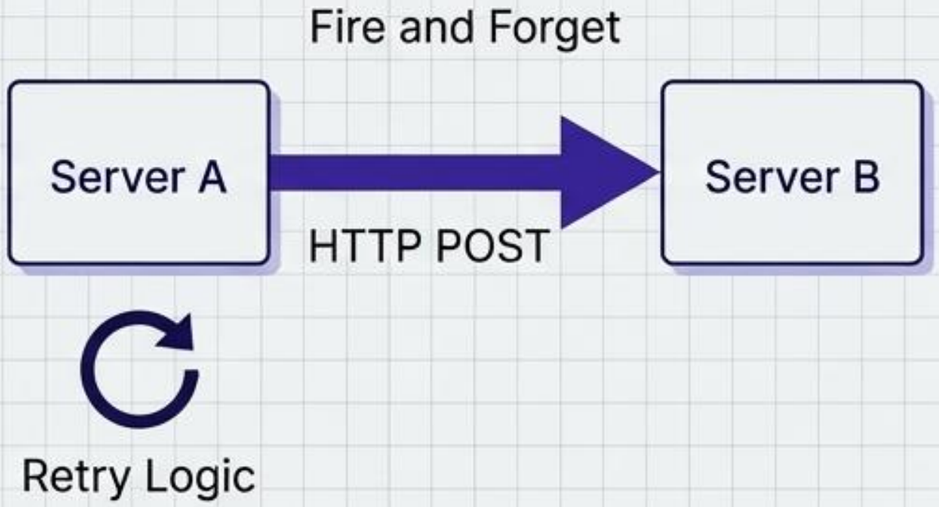
Webhooks: The Event-Driven Trigger

Protocol Anatomy

Latency:	Hundreds of ms to Seconds
Direction:	One-Way
Browser Native:	No



Key Traits

- Simplest integration pattern: expose a URL.
- Strictly event-driven server-to-server communication.
- Receiver cannot push back.

Ideal Use Cases

- Payment confirmations (Stripe)
- CI/CD pipeline triggers
- CMS content updates

Limitation: Not for frontends. Requires an intermediary server.

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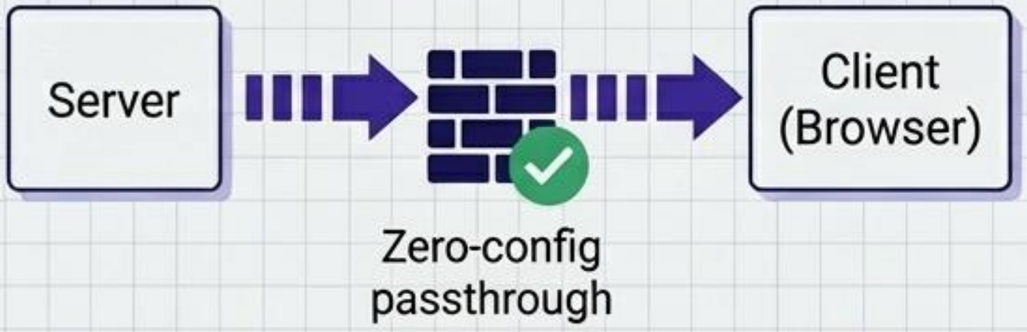
Server-Sent Events (SSE): The Unidirectional Stream

Protocol Anatomy	
Latency:	Seconds
Direction:	One-Way
Browser Native:	Yes (EventSource)



Key Traits

- Standard HTTP—seamlessly traverses proxies and CDNs.
- Text-only payload constraints.
- Auto-reconnects natively via Last-Event-ID.

Ideal Use Cases

- AI token streaming (ChatGPT)
- Live analytical dashboards
- Notification feeds

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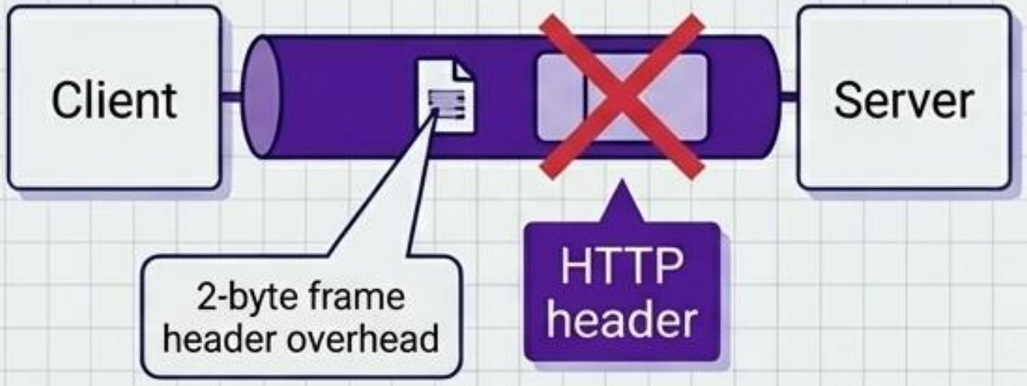
WebSockets: The Bidirectional Backbone

Protocol Anatomy

Latency:	Tens of ms
Direction:	Bidirectional
Browser Native:	Yes



Key Traits

- Persistent, stateful connection.
- Full-duplex TCP over a single HTTP upgrade.
- Both sides can push data asynchronously.

Ideal Use Cases

- Realtime chat applications
- Multiplayer gaming
- Collaborative editors

Limitation: Lacks native NAT traversal; cannot handle UDP media streams.

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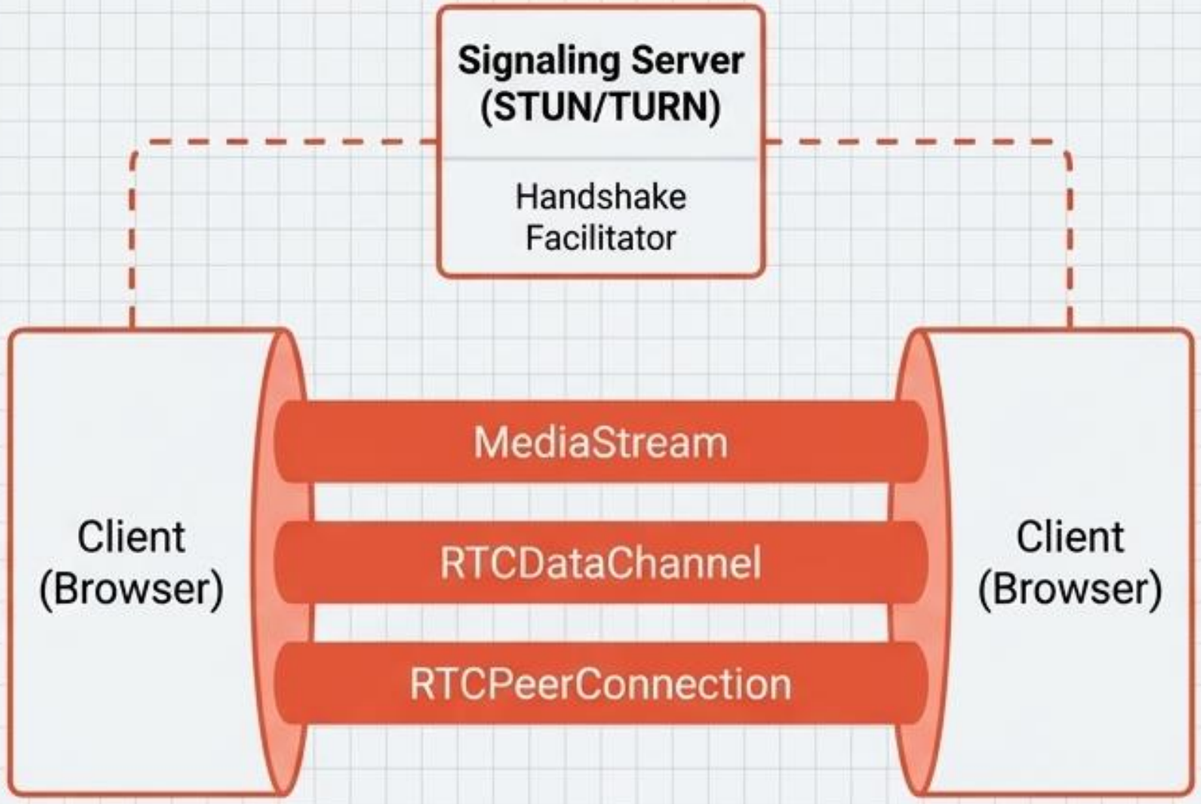
WebRTC: Sub-Millisecond Peer-to-Peer

Protocol Anatomy

Latency:
Sub-ms to ms

Direction:
Peer-to-Peer

Browser Native:
Yes (No plugins)



Key Traits

- Bypasses server chokepoints after initial signaling.
- Supports audio, video, and arbitrary data over UDP.
- Massive API surface area.

Ideal Use Cases

- Video conferencing
- Screen sharing
- Ultra-low latency gaming

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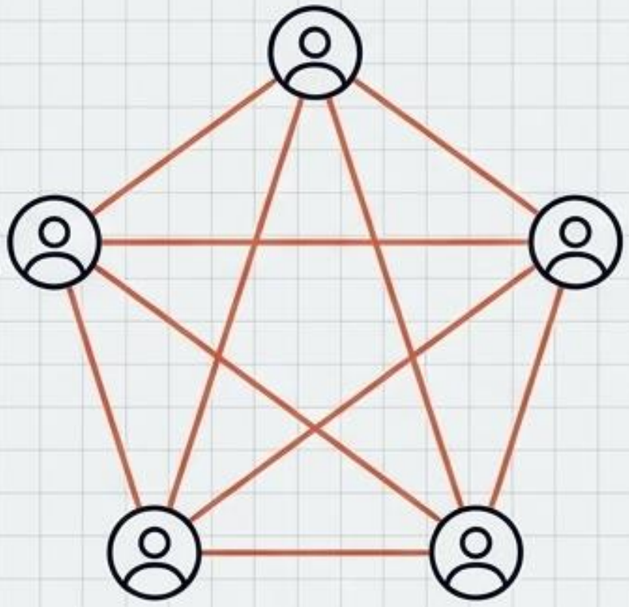
The WebRTC Scaling Bottleneck

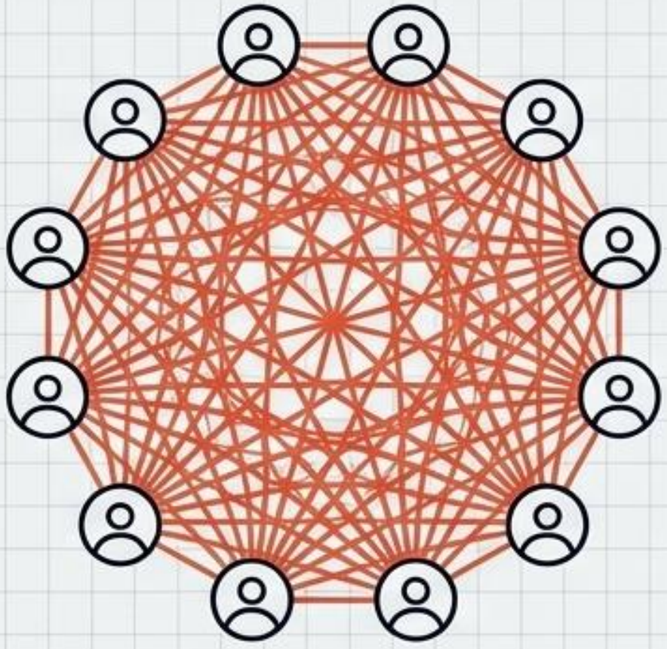
Pure peer-to-peer architecture falls apart under group scale due to exponential connection growth.

Manageable Scale



Users (N) = 5
Total Connections = 10

System Failure



Users (N) = 10
Total Connections = 45

N*(N-1)/2

With just 10 participants, client CPUs choke on encoding and decoding 45 concurrent streams.

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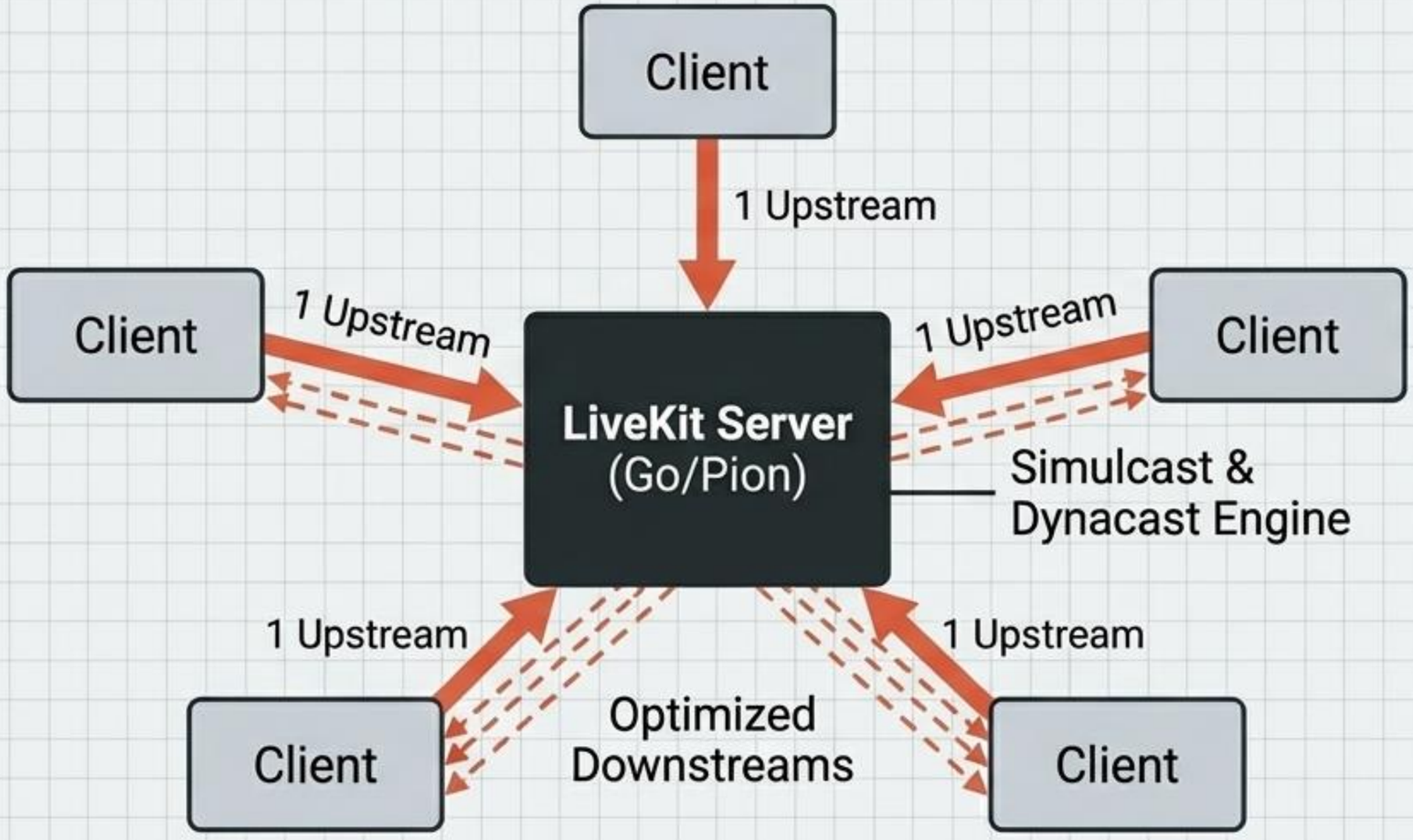
Selective Forwarding Units (SFUs) & LiveKit

LiveKit resolves the WebRTC mesh bottleneck by replacing peer-to-peer chaos with an intelligent, central routing node.



Capabilities

- Scales to hundreds of users per room.
- Built-in speaker detection and stream recording.
- Cross-platform SDKs (React, iOS, Unity).
- Native AI Agents framework.

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The Protocol Matrix

Protocol	Transport	Direction	Latency	Complexity	Ideal Use Case
Webhooks	HTTP POST	Server -> Server	Seconds	Low	System Triggers
SSE	HTTP	Server -> Client	Seconds	Very Low	Feed Updates
WebSockets	TCP	Bidirectional	Tens of ms	Medium	Chat / Collab
WebTransport	UDP (HTTP/3)	Bidirectional	Tens of ms	High	Next-Gen Fast State
WebRTC	UDP	Peer-to-Peer	Sub-ms	Very High	Live Media

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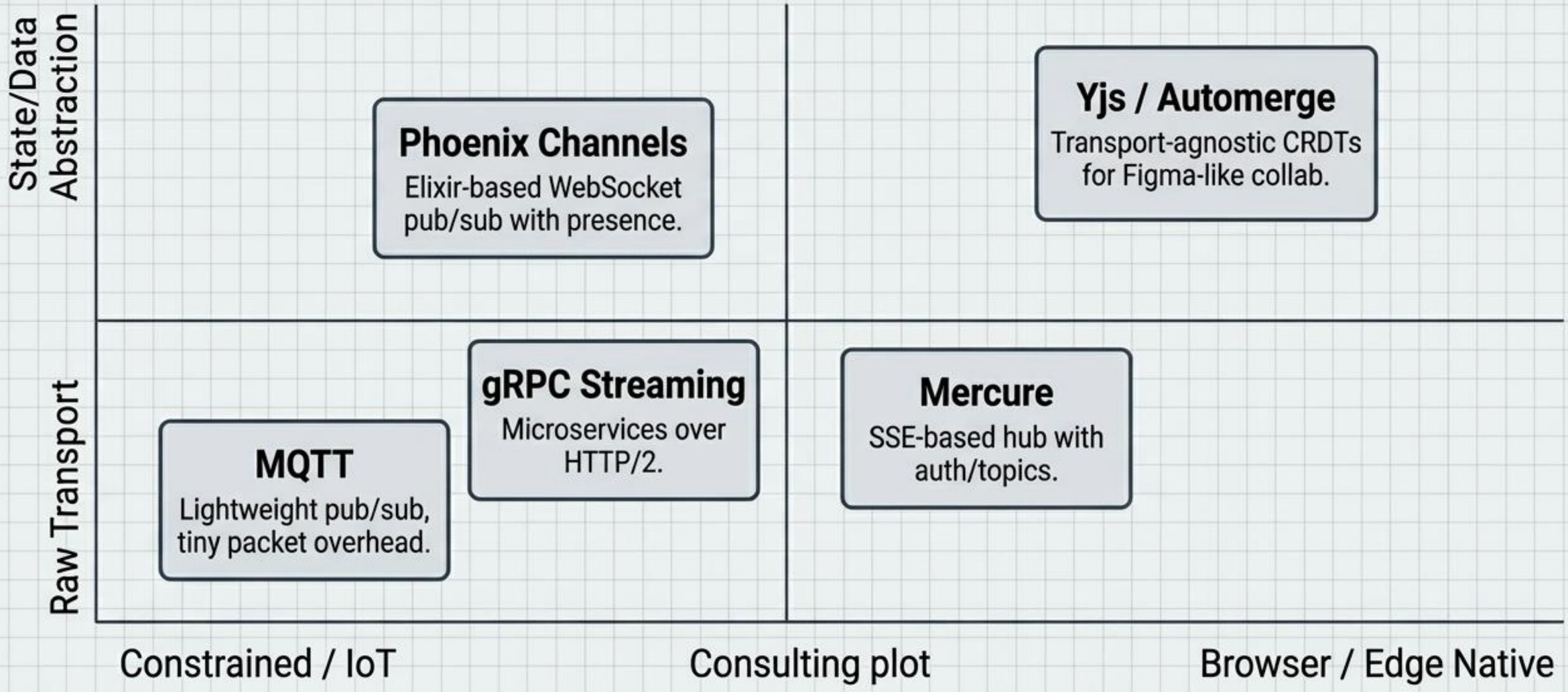
Mapping the Broader Ecosystem



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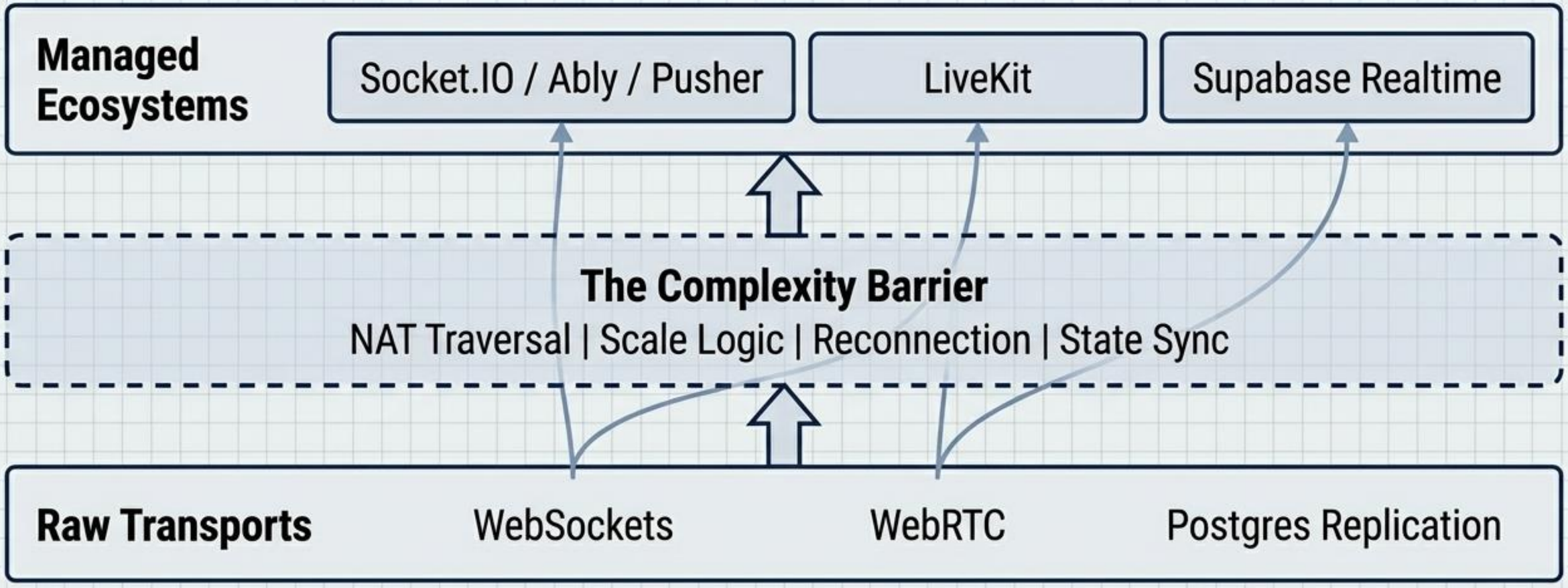
The Abstraction Layer

Modern infrastructure encapsulates scaling, fallback logic, and presence tracking, isolating developers from raw protocol friction.



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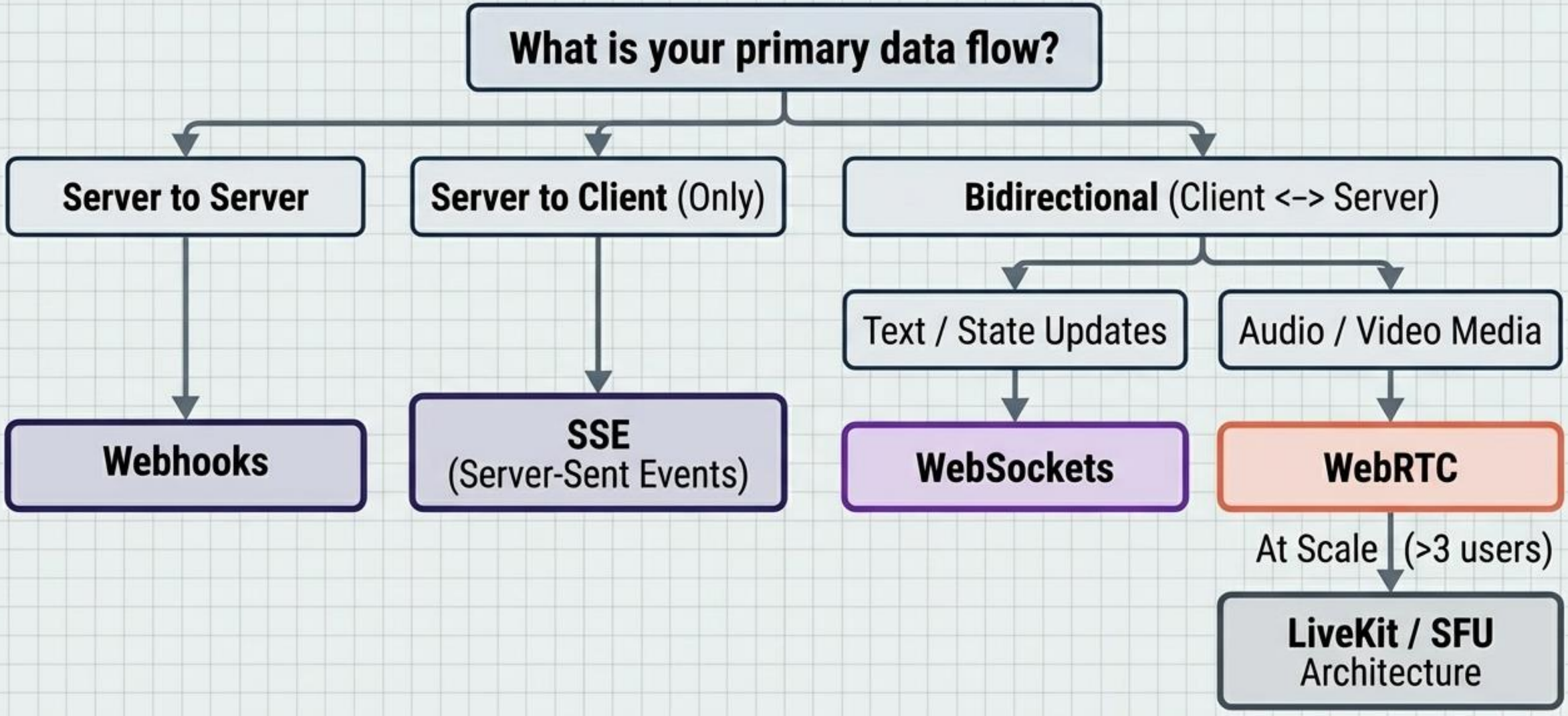
The Realtime Routing Tree



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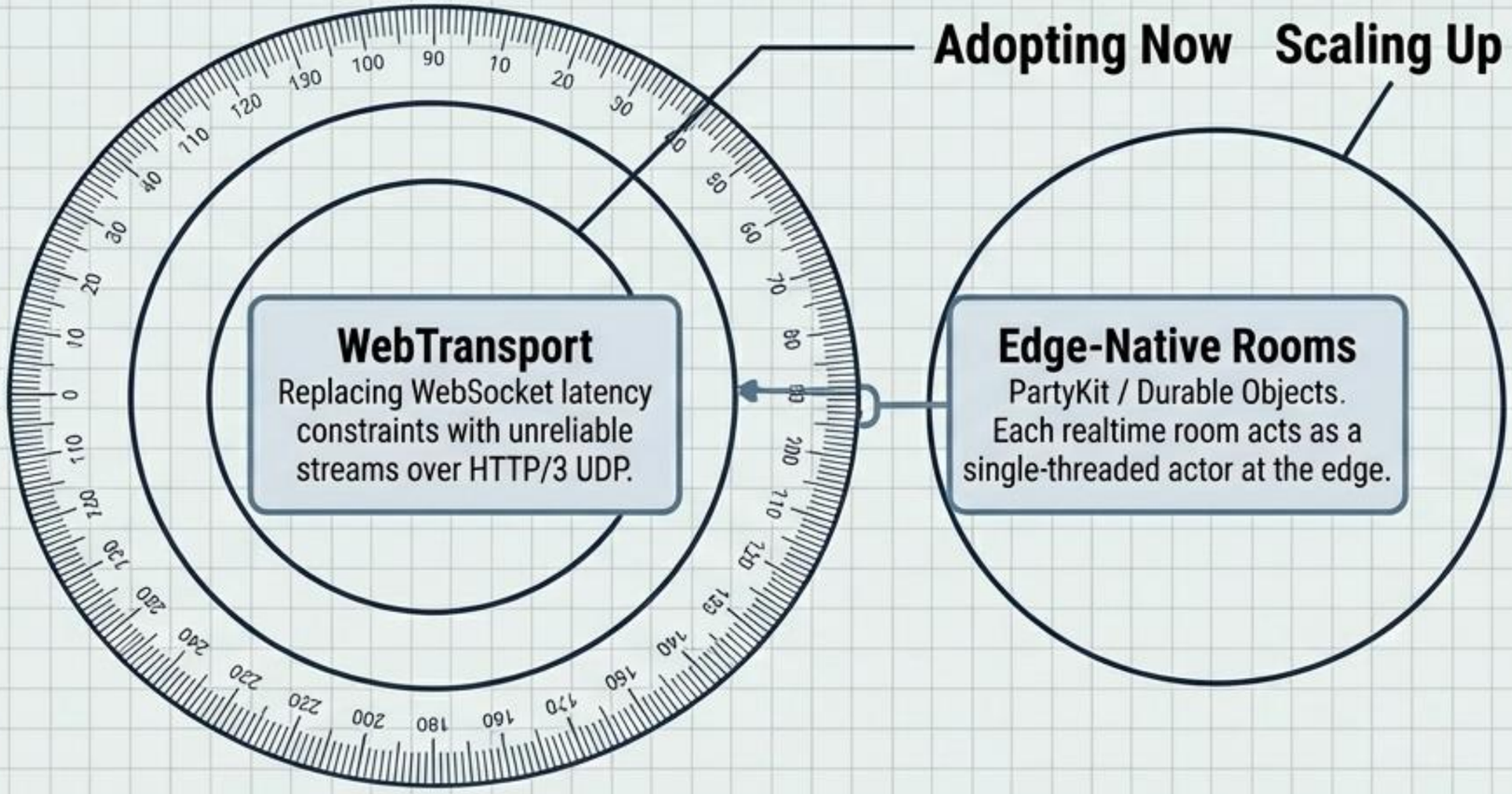
The Realtime Frontier



Adopting Now

WebTransport
Replacing WebSocket latency constraints with unreliable streams over HTTP/3 UDP.

Scaling Up

Edge-Native Rooms
PartyKit / Durable Objects. Each realtime room acts as a single-threaded actor at the edge.

Bleeding Edge

AI Realtime Participation

LiveKit Agents. Server-side AI bots natively joining WebRTC rooms for ultra-low latency voice.

The next era of realtime isn't just about moving data faster; it's about pushing state to the edge and inviting AI into the stream.

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